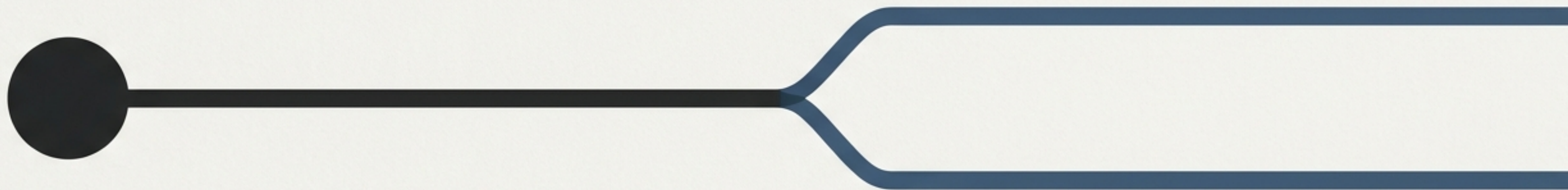


Diagnostic Coordinates and the Time of Repair

Correcting the Anti-Alignment Theorem



Based on the 2026 paper by Flyxion

The Two-Class Myth of System Repairability

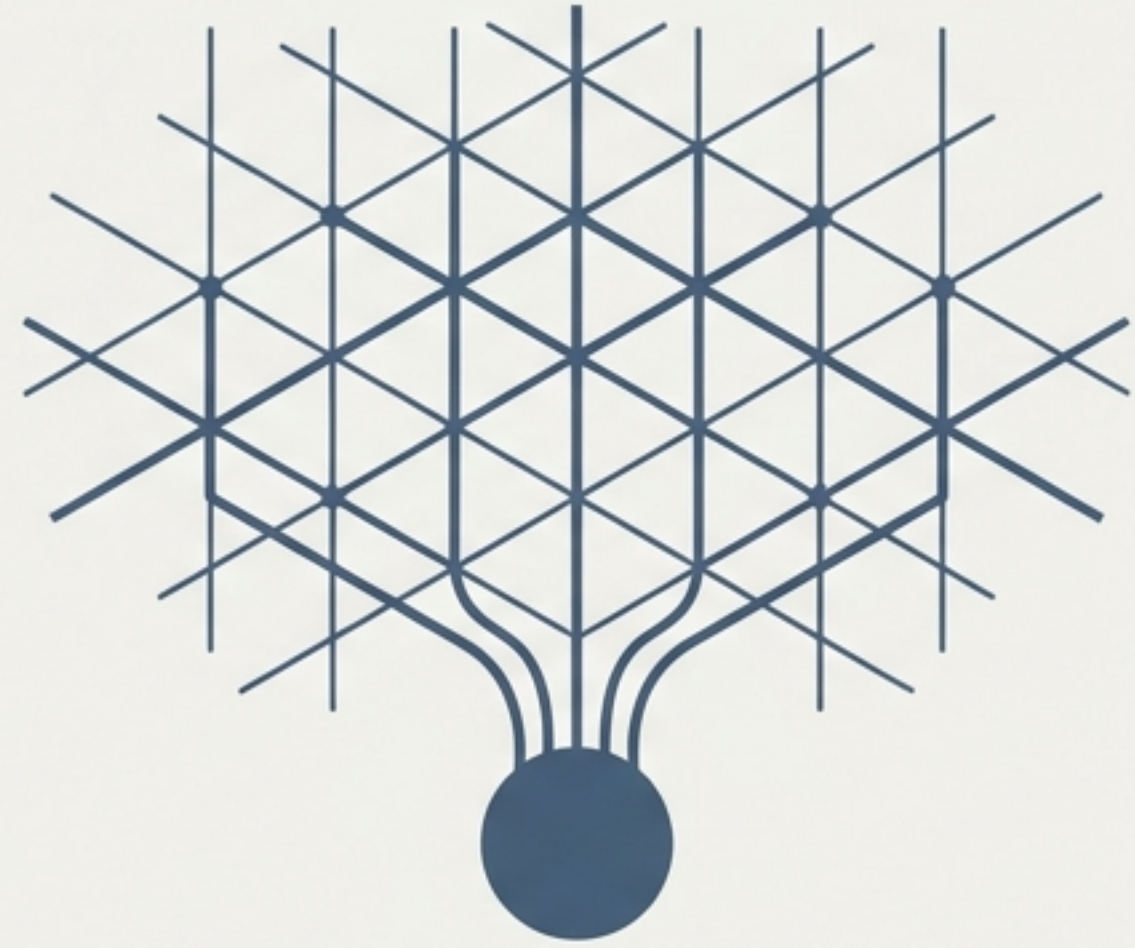
History-First Origin



A Folk Recipe

- Optimized for execution, pathological for diagnosis.

Theory-First Origin



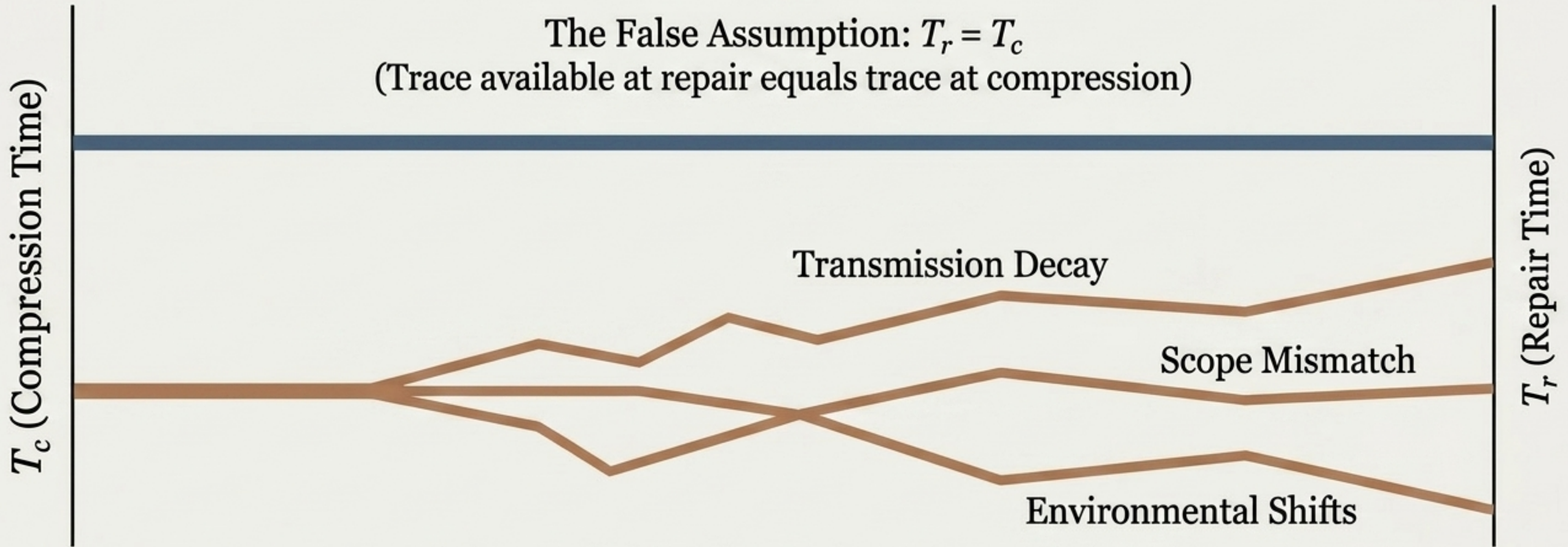
Engineered Codebase

- Explicit derivations assumed to guarantee repairability.

The Flawed Corollary: The Anti-Alignment Theorem assumes systems born from trial-and-error are inherently repair-poor from birth, while engineered systems indefinitely retain their diagnostic derivations.

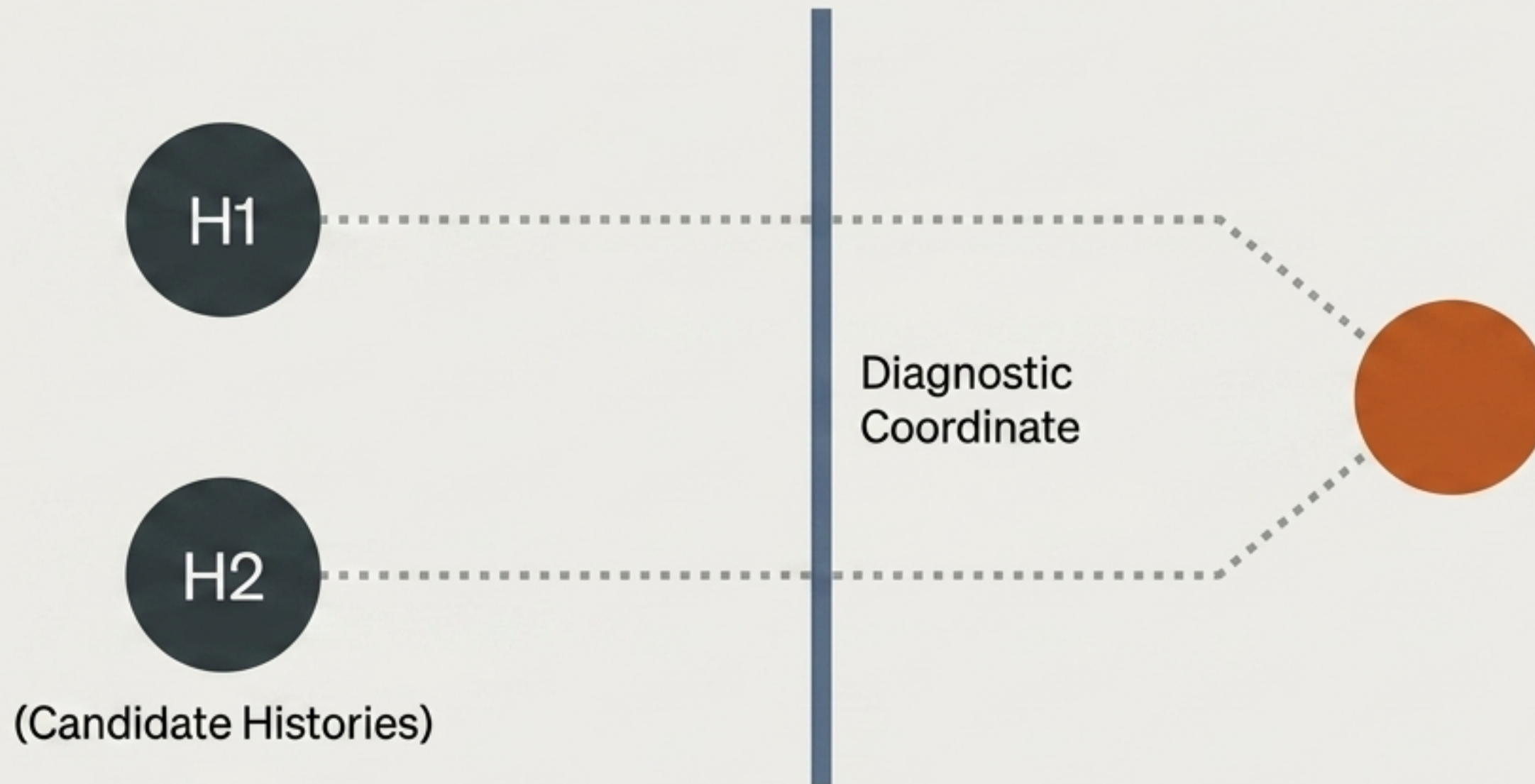
The Hidden Flaw: The Synchrony Assumption

The original theorem assumes an operator's repairability is fixed permanently at birth, failing to account for what happens to diagnostic information after compression.



The Reality: Information diverges, degrades, or loses context.

Definition 3.1



The Definition

A retained distinction between two otherwise indistinguishable candidate histories that produce different present failures.

The Metric

$D(O, t)$ is the number of diagnostic coordinates retained and accessible for operator O at time t .

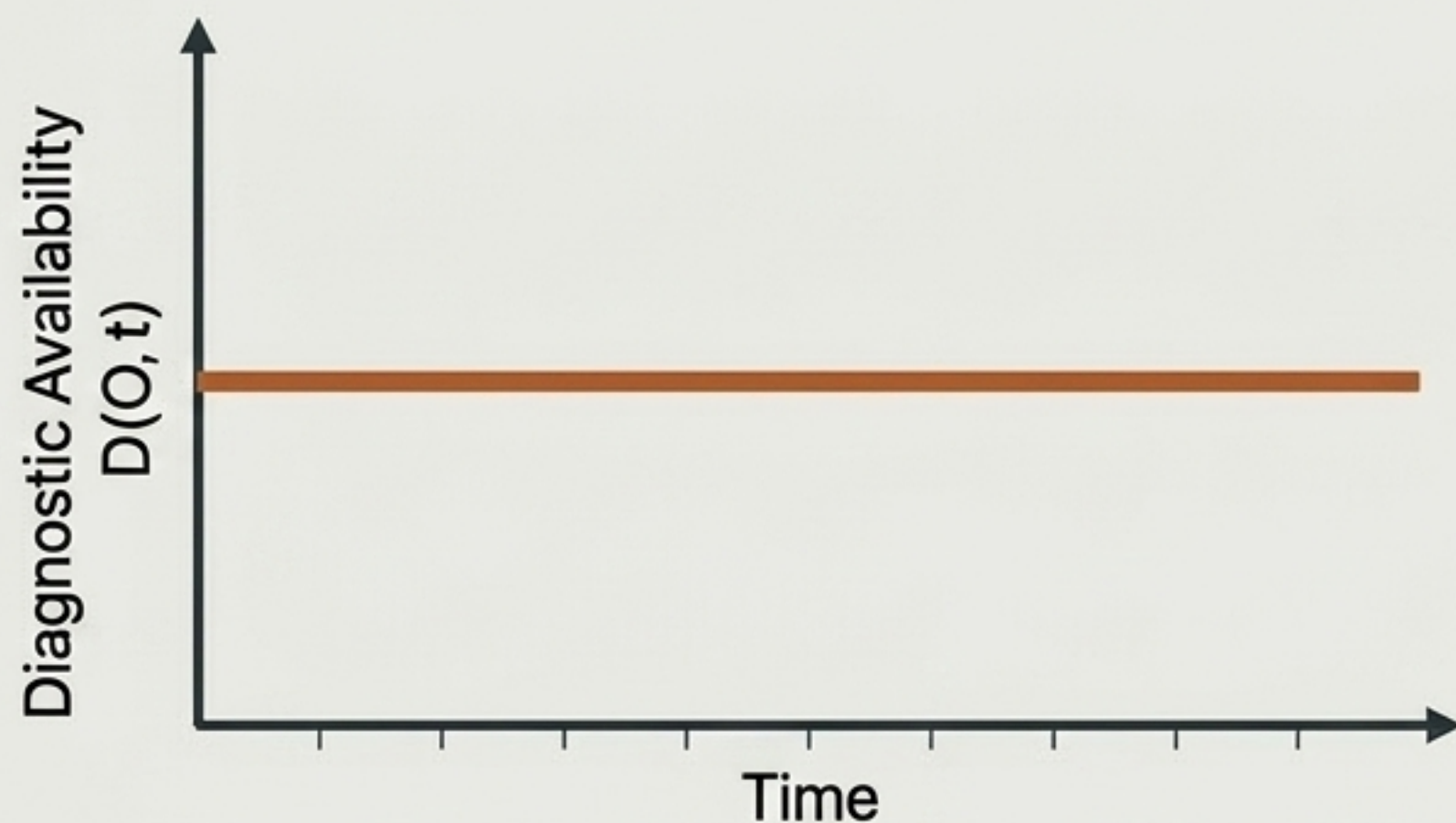
Origin Does Not Predict Outcome

	History-First Origin	Theory-First Origin
Assumed Repairability (Old Paradigm)	Low	High
Actual Repairability at Time t (New Paradigm)	Low	Low (Subject to post-design coordinate loss)

Key Insight: A theory-first operator can end up exactly as repair-poor as a history-first one. Repairability is defined by what happens to diagnostic coordinates after compression, not how they were born.

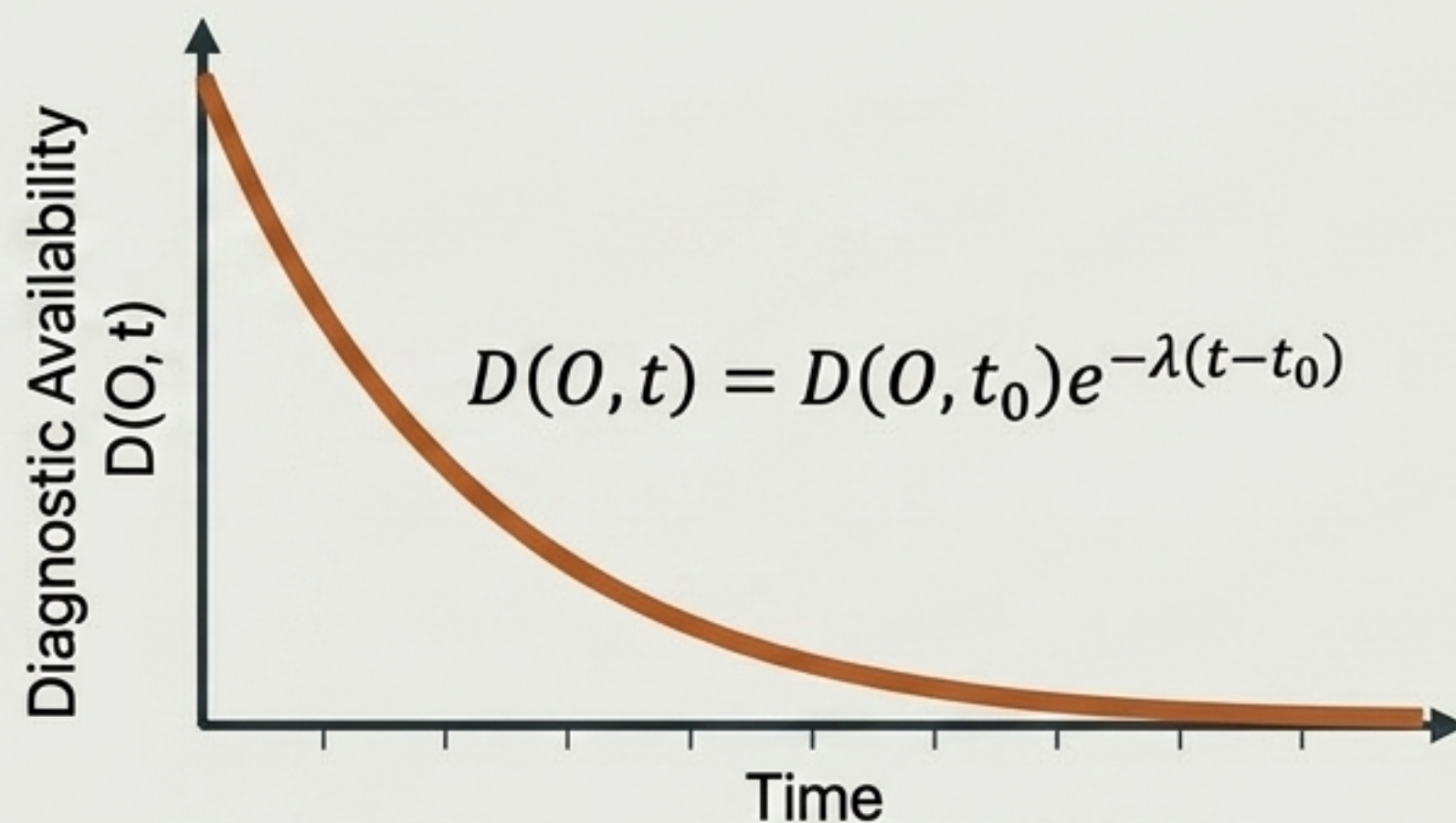
Two Paths to Identical Diagnostic Poverty

Case 1: The Altitude Recipe (History-First)



The recipe fails at altitude because discarded distinctions (e.g., atmospheric pressure) were never retained at compression.

Case 2: The Legacy Codebase (Theory-First)

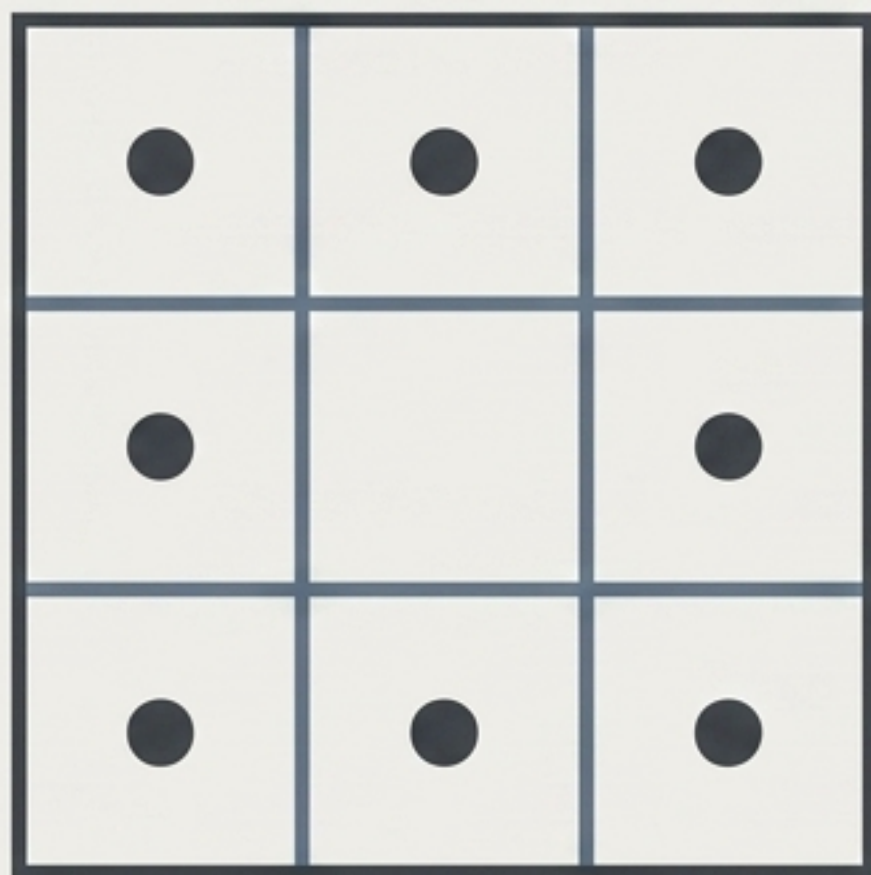


The codebase fails because coordinates present at design time decayed via personnel turnover ($\lambda > 0$). No fixed advantage survives unbounded decay.

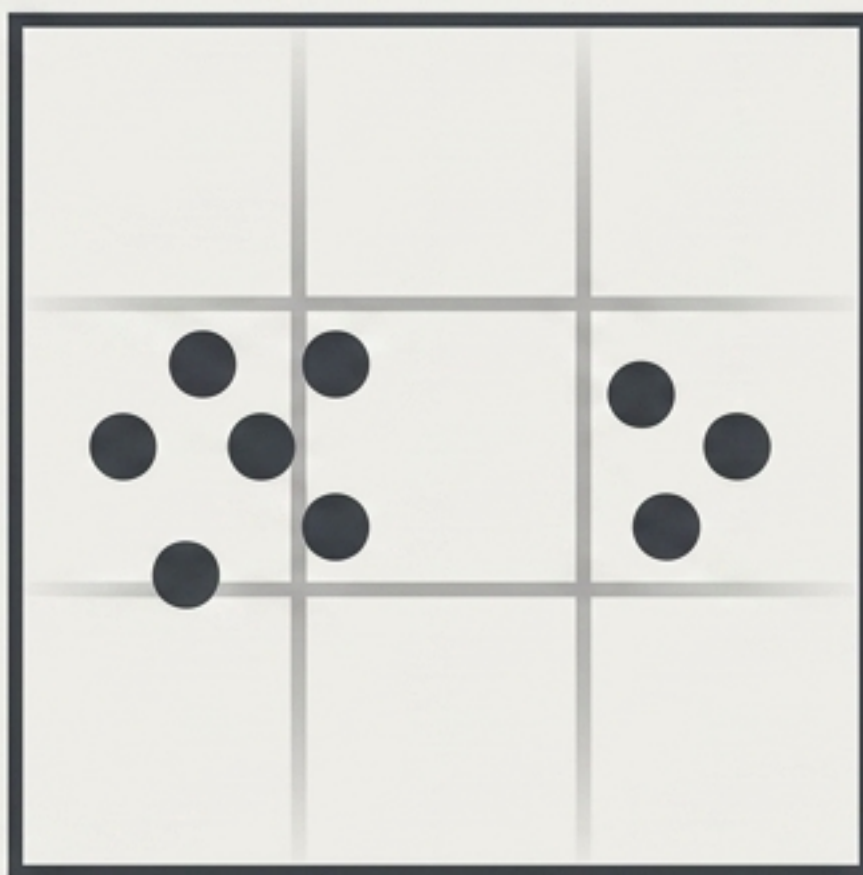
The Mathematics of Forgetting: Partition Coarsening

$$R(s \mid T) = \log |\{h \in H : h \sim_T O, h \vdash s\}|$$

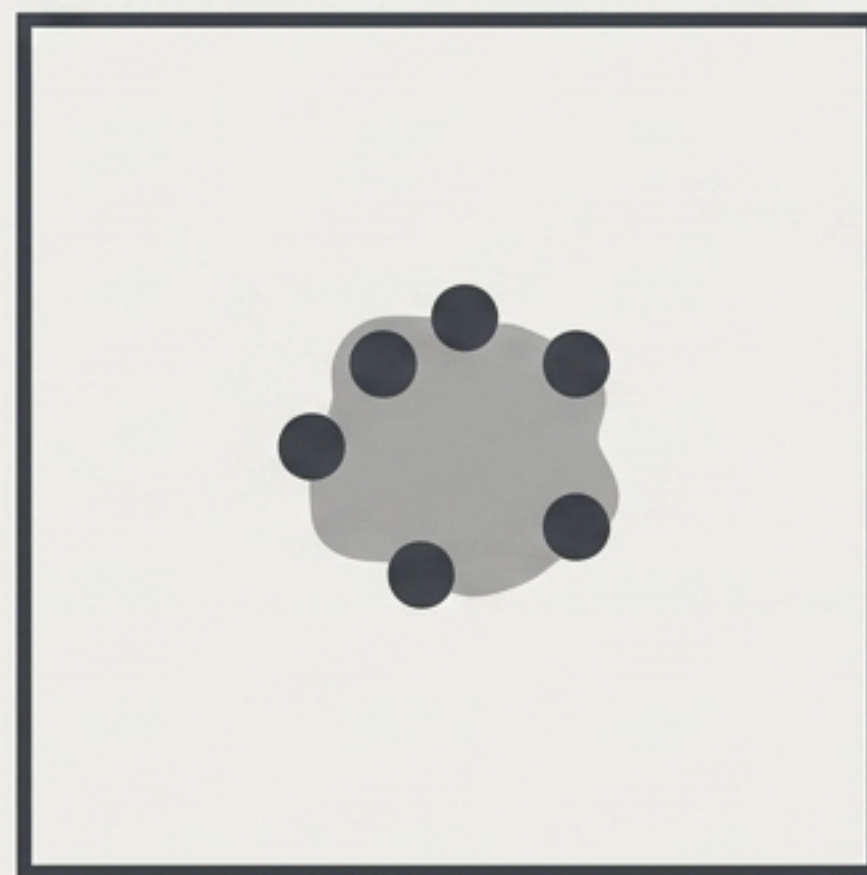
High D (Full Coordinates)



Decay (Coarsening Partitions)



Low D (Maximum Entropy)



Core Principle: Deleting a diagnostic coordinate merges previously distinguished histories. As partitions coarsen, Repair Entropy strictly increases.

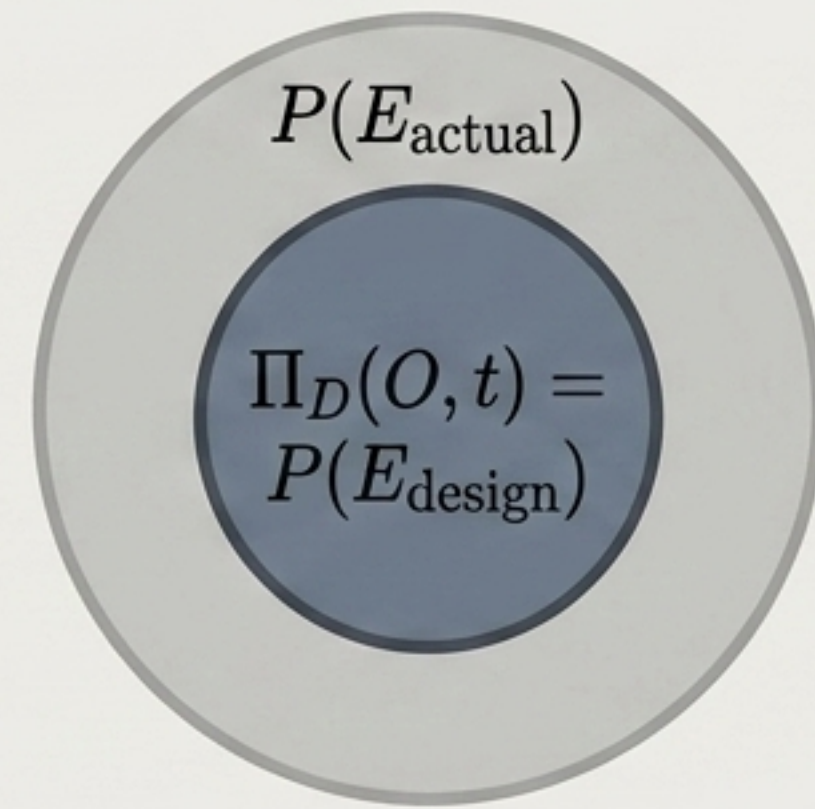
Case 3: The Danger of Fault-Space Dependence

Operational Insufficiency (Safe Refusal)



Refusal to run is safe. The system halts immediately when required operational coordinates are absent.

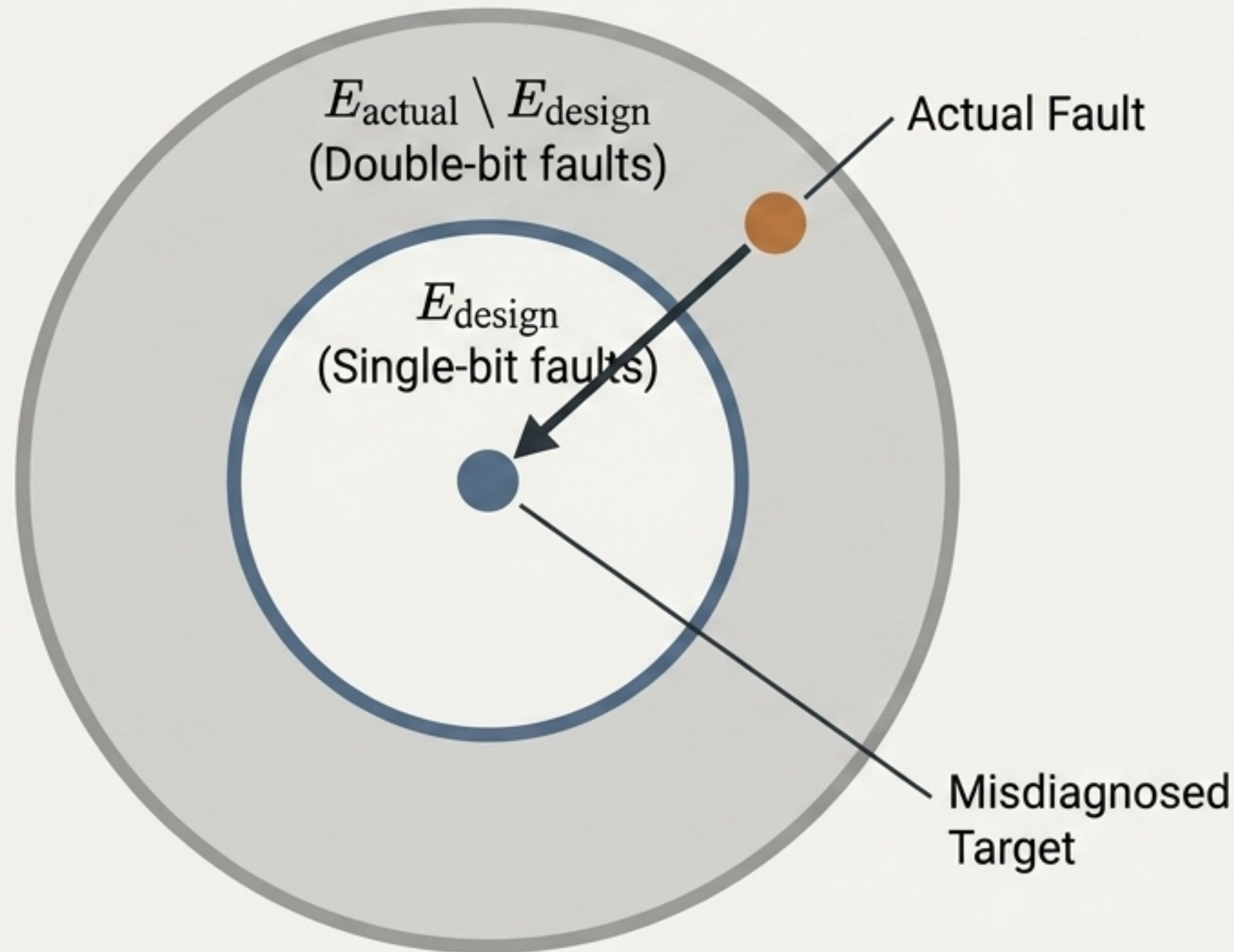
Scoped Diagnostic Sufficiency (Scope Mismatch)



Scope mismatch is dangerous. A Hamming(7,4) decoder has coordinates mechanically checkable for single-bit faults (E_{design}). But it is blind to the outer ring of double-bit faults (E_{actual}).

Confident Error: The Illusion of Successful Repair

Confident Error Radar



1 The Mechanism

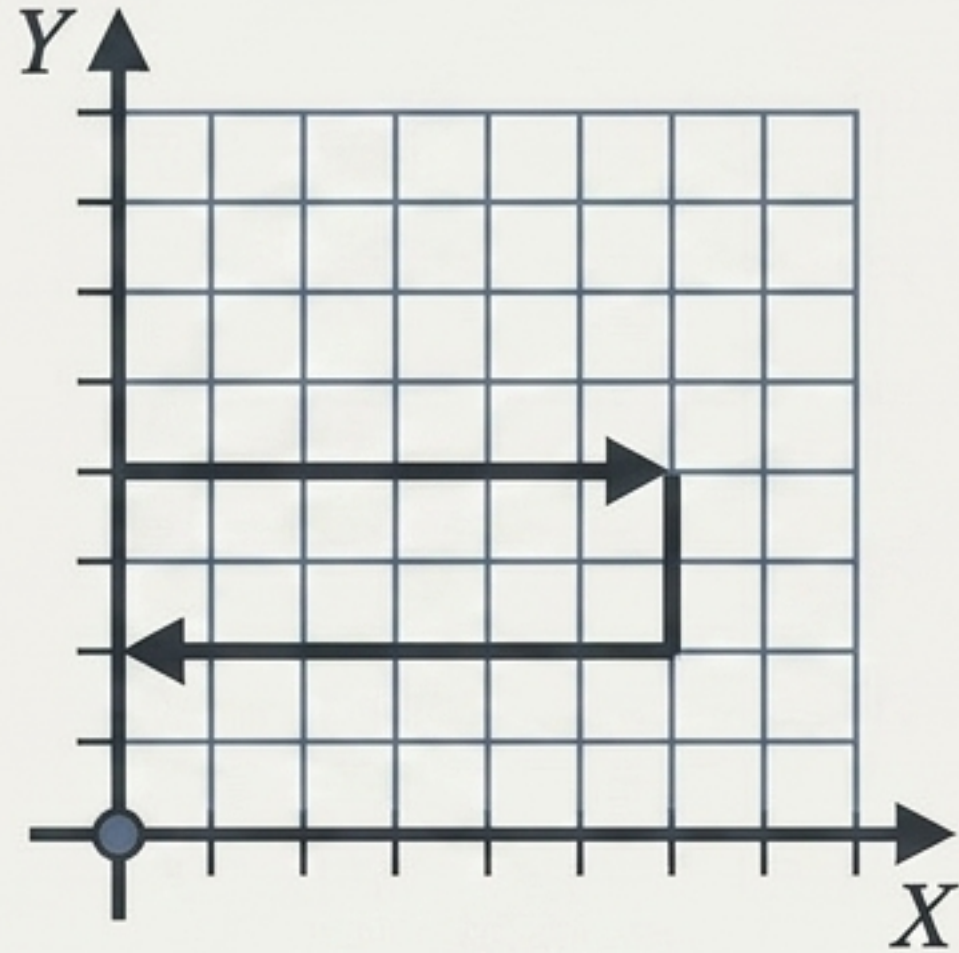
The decoder computes a syndrome $\sigma(e_1) = \sigma(e_2)$. It deterministically interprets the out-of-scope fault as evidence for an in-scope fault.

2 The Result

The system does not refuse the repair. It repairs the wrong location with full confidence, successfully passing all internal admissibility tests.

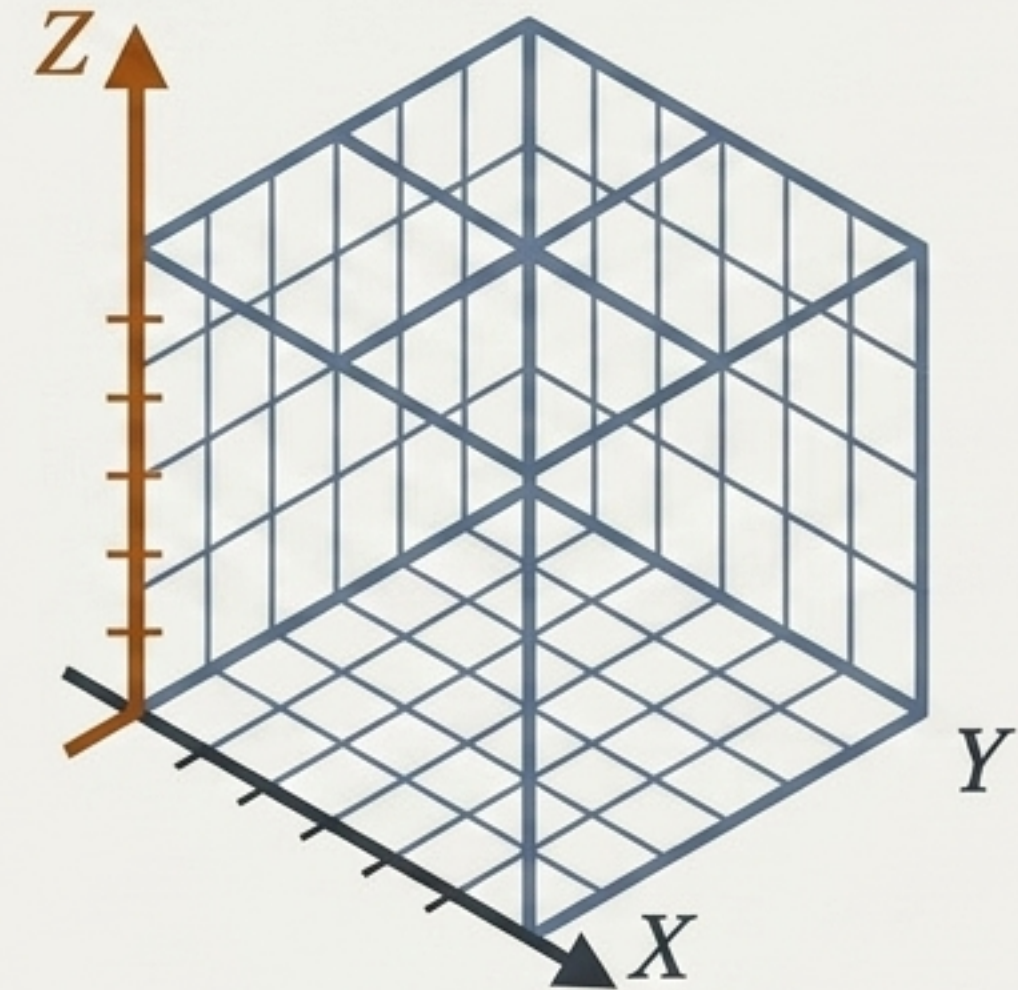
Vertical vs. Horizontal Repair Interventions

Vertical Repair



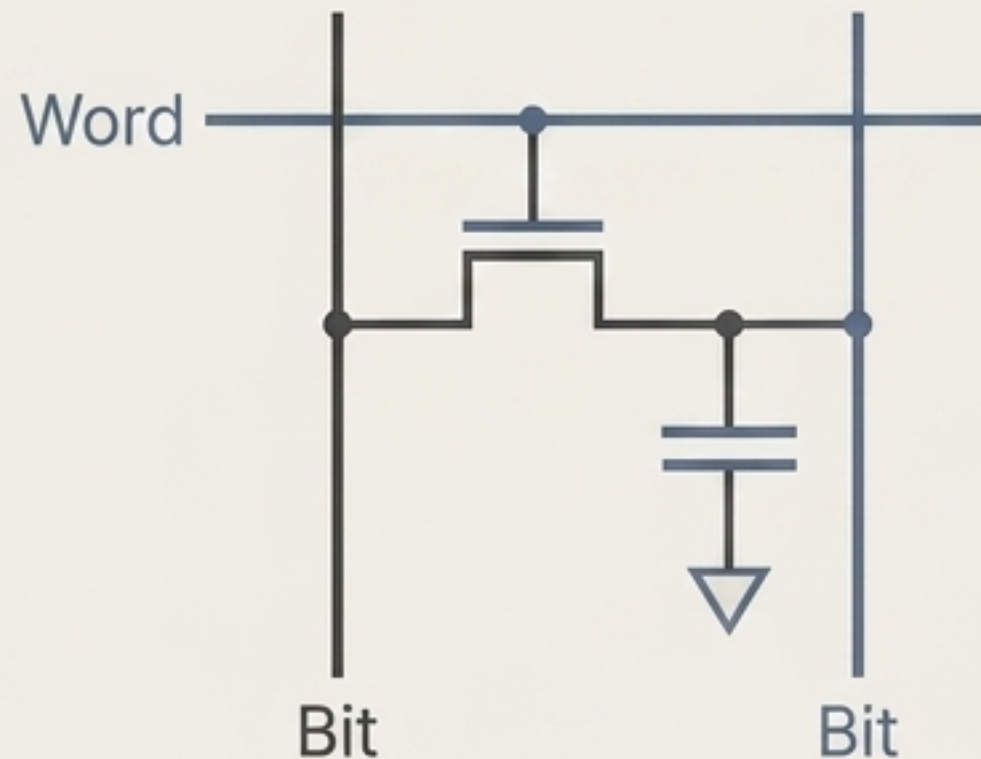
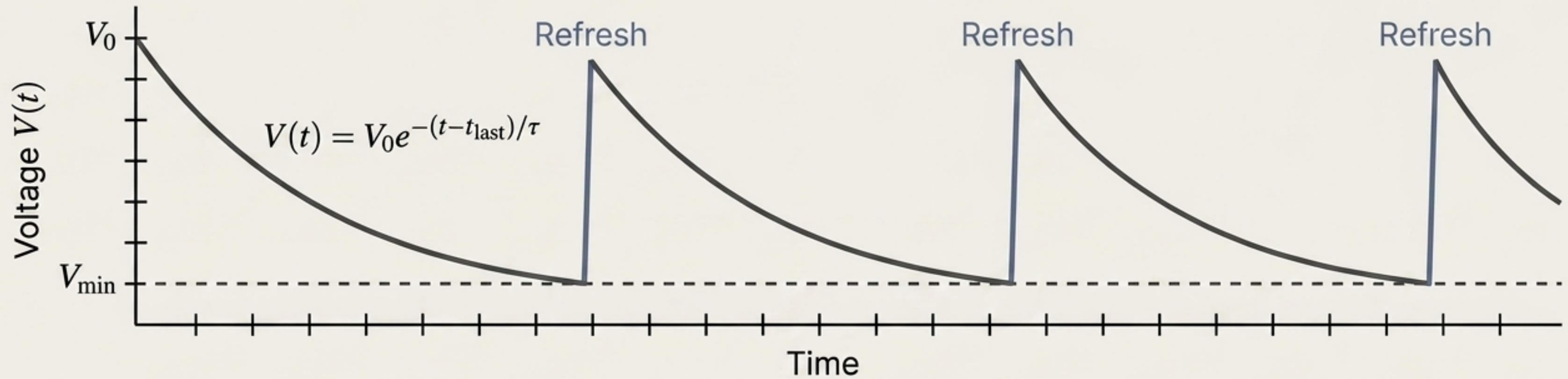
Vertical Repair operates entirely within a fixed diagnostic vocabulary $\mathcal{C}(\mathcal{O}, t)$. It recovers lost coordinates of an already known type.

Horizontal Repair



Horizontal Repair requires $\mathcal{C}(\mathcal{O}, t') \supset \mathcal{C}(\mathcal{O}, t)$. It introduces a fundamentally new coordinate type (e.g., atmospheric pressure for the recipe) that the original vocabulary could not express.

Case 4: Active Regeneration in DRAM



The Concept

Left alone, a memory cell's readability follows the exact same passive decay trajectory as the legacy codebase.

The Solution

Continual reinvestment. Refresh rewrites the cell at intervals $\Delta t < \tau \log(V_0/V_{\min})$, maintaining the coordinate condition indefinitely.

The Four Mechanisms of Diagnostic Failure

System Component	Mechanism of Failure	Repair Intervention Required
Altitude Recipe	Never retained	Horizontal Redesign
Legacy Codebase	Passive decay over time (t)	Vertical Recovery
Hamming Decoder	Scope mismatch (E)	Preventative Redundancy (SECDED)
DRAM Cell	Prevented decay via active refresh	Scheduled Maintenance

What began as a single temporal correction to the Anti-Alignment Theorem resolves into four distinct mechanisms. Origin explains none of them on its own.

The Unified Framework of Repairability

The diagram illustrates the equation $D(O, t; E) = f(C(O, t), A(O, t), E)$ with several annotations. Above the first parameter O is the text "Diagnostic Coordinate Availability", with a bracket underneath it. Above the function f is the text "Representable Vocabulary (The 2D vs 3D space)", with a bracket underneath it. Below the parameter $A(O, t)$ is the text "Temporal Availability (Decay vs Refresh)", with a bracket underneath it. Below the parameter E is the text "Fault Space (The scope boundary)", with a bracket underneath it.

$$D(O, t; E) = f(C(O, t), A(O, t), E)$$

Diagnostic Coordinate Availability

Representable Vocabulary
(The 2D vs 3D space)

Temporal Availability
(Decay vs Refresh)

Fault Space
(The scope boundary)

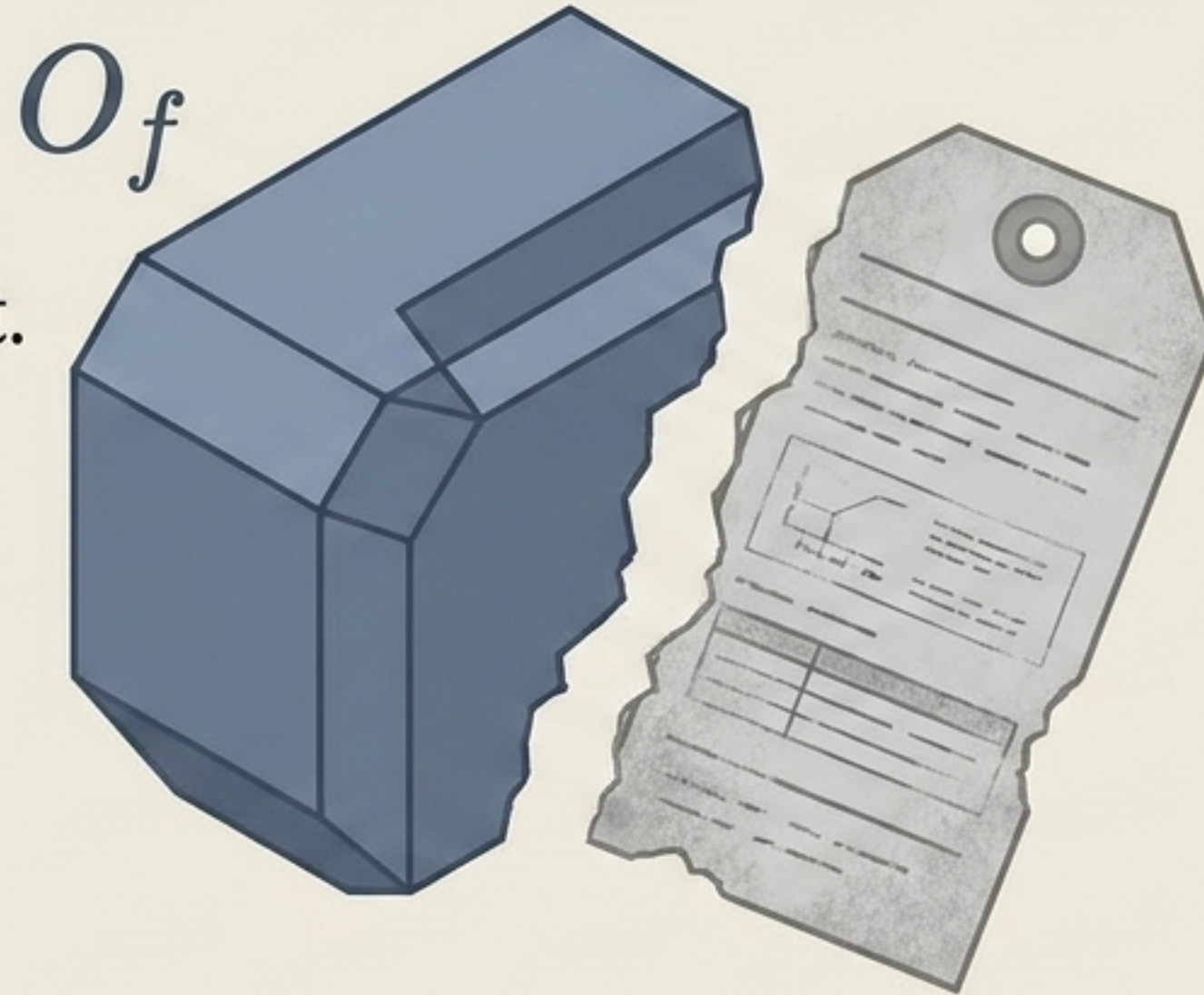
Core Insight: Repairability is not a static property fixed at birth. It is a dynamic function of the system's evolving vocabulary, its maintenance through time, and the specific fault space it is forced to confront.

The Final Test: Isolating Origin from Availability

The Stripped Component.

The Experiment

Take a perfect, theory-first engineered component. Deliberately strip all documentation and derivations before shipping, holding the physical operator unchanged.



The Hanging Question

If repairability drops to match the trial-and-error recipe, $D(O, t)$ is proven as the sole operative variable. The test remains unrun.